IPCON® TPU CF 70D

Carbon Fiber Reinforced Polyurethane 70D Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	380 ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.09 g/cm³
Shrinkage Rate*	Specially-defined	0.13% - 0.42%
Surface Hardness	Shore hardness	~ 70D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	1.16%
Melt Index	220 °C, 2.16 kg	3.5 g/10 min
Melting Temperature	DSC, 10 °C/min	202 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	N / A
Heat Deflection Temperature	ISO 75, 0.45 MPa	N / A

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	27 MPa
Tensile Strength Z	ISO 527, GB/T 1040	17 MPa
Young's Modulus XY	ISO 527, GB/T 1040	773 MPa
Young's Modulus Z	ISO 527, GB/T 1040	495 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	38.4%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	48.7%
Flexural Strength XY	ISO 178, GB/T 9341	22 MPa (No Break)
Flexural Strength Z	ISO 178, GB/T 9341	14 MPa (No Break)
Flexural Modulus XY	ISO 178, GB/T 9341	427 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	261 MPa
Impact Strength XY	ISO 179, GB/T 1043	> 124.0 kJ/m² (No Break)
Impact Strength Z	ISO 179, GB/T 1043	41.4 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 230 °C, bed temperature: 45 °C, printing speed: XY—147 mm/s and Z—70 mm/s) and IPCON TPU CF 70D (Black) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill

pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Impact and fall-resistant, wear-resistant and shock-absorbing, durable, anti-freeze and anti-bacterial, water and oil-resistant, less strings, high-speed and easy to print, suitable for mainstream printers, and compatible with AMS and CFS.
2. Can be used to print toy racing cars and drone accessories, robot dog shells, mechanical skins, children's footballs, refrigerator egg holders, swimming pool pipe fittings, kitchen draining tools, flexible bar mats and other items.

• Recommended Printing Settings

Drying Settings Before Printing	80 - 100 °C, 8 - 12 h (Blast drying oven)
Printing Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself, PLA or support filament for TPU
Compatible Printer Type	Open-frame, enclosed-frame (Suggest to open the front door and remove the upper lad)
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plates
Plate Surface Preparation	For most kinds of plate, applying glue is not needed. If the first layer bonding is not firm enough, apply liquid glue or PVP solid glue stick.
Nozzle Temperature	210 - 250 °C
Bed Temperature	35 - 50 °C
Fan Speed	10% - 90%
Max Print Speed	250 mm/s (P1S, P2S, 230 °C) (For the thin and high models and the support structure, the recommended speed is lower than 150 mm/s)

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